Labour Demand 1: Competitive Labour Markets

EC306- Wages and Employment

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Goals of This Section

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- Understand how firms decide how much labour to employ to produce
- Compare short-run vs. long-run labour demand decisions
- Explain why labour demand is downward sloping
- Learn factors that affect the elasticity of labour demand
- Understand the impact of free trade and international competition on the Canadian labour market

Background

Background

- Labour demand refers to the number of workers that firms want to hire at a given wage rate
- It is a **derived demand**
  - Linked to demand for the goods/services produced
  - It is an input into that production
- Several factors will influence labour demand
  - Firm’s objectives and constraints
  - Demand conditions in product markets
  - State of technology
  - Time horizon
    - **Short run:** one or more inputs fixed (usually capital)
    - **Long run:** all inputs variable

Structure of Product Markets

- We noted that labour demand is driven by the firm's output
  - Labour is used as a input
- The structure of the product market will influence the demand for labour
  - Ranges from perfect competition (many sellers) to monopoly (one seller)
- By contrast, the structure of the labour market will influence the supply the firm faces
  - Ranges from perfect competition (many workers) to monopsony (one buyer)
- In this section we examine the case of perfect competition in both product and labour markets

Demand in the Short Run

Model

- As labour demand is derived from production, the model starts with a production function
  - Relates inputs to outputs

$$Q = f(K, N)$$

- Where
  - Q is quantity of output
  - K is capital (fixed in short run at K_0)
  - N is labour (variable in short run)
- Firms objective is to maximize profit
- In the short run this means choosing N given fixed K_0
- Optimality is where the marginal revenue product of labour equals its price

Model

- To see this mathematically, take output prices are taken as given (perfect competition).
- Profit

$$\pi(N) = pQ - wN - rK_0$$

$$= pF(K, N) - wN - rK_0$$

- Differentiate $\pi(N)$ w.r.t. N:

$$\frac{d\pi}{dN} = p\frac{\partial F}{\partial N}(K_0, N) - w$$

$$= pMP_N - w$$

Model

- Optimal N^* satisfies

$$pMP_N = w$$

- Where
  - MP_N is the marginal product of labour
  - pMP_N is the marginal revenue product of labour (MRP)
    - When firm is in perfectly competitive product market and $MR = p$ it is called value of marginal product (VMP)
  - w is the wage rate (marginal cost of labour, MC_N)
- Optimality requires that the value of the last unit of labour hired equals its cost

Model

- Other key metrics include
  - Total Revenue Product(TRP): Total revenue associated with labour employed

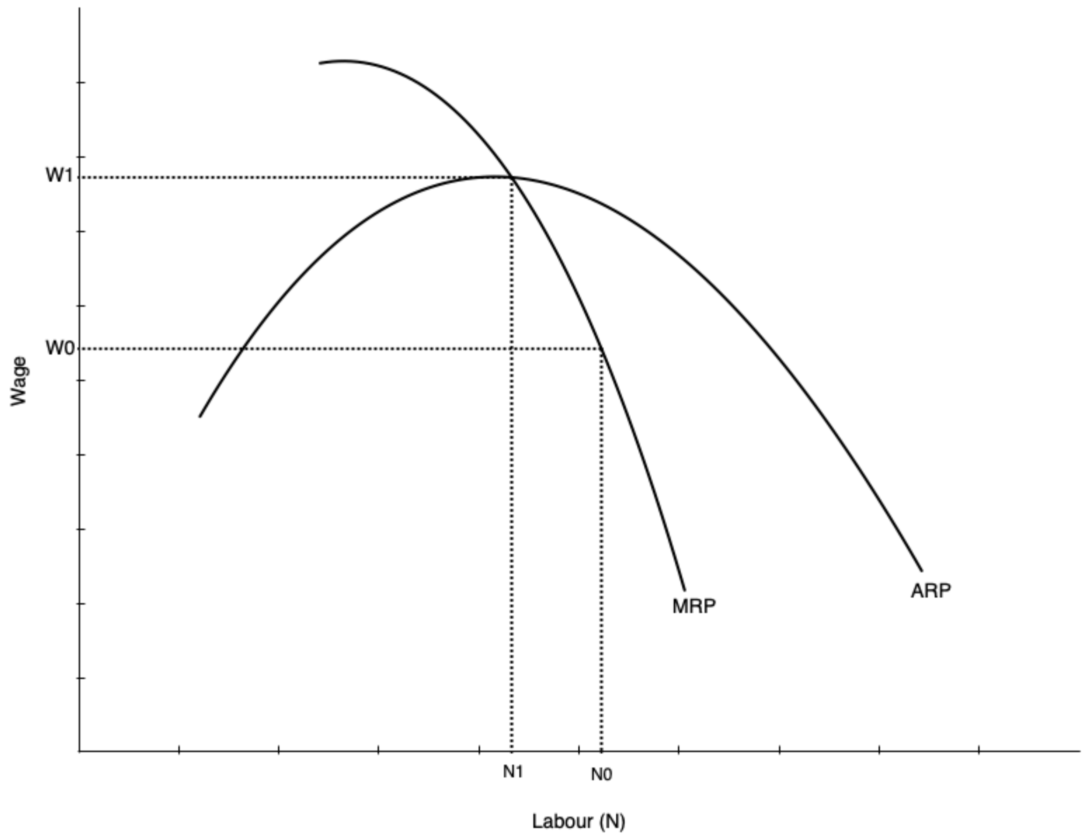
$$TRP = p \times Q$$

  - Average Revenue Product(ARP): Average revenue per unit of labour employed

$$ARP = \frac{TRP}{N} = \frac{p \times Q}{N}$$

- As noted, the firm's optimal choice of labour is where $MRP_N = w$
  - This defines its labour demand curve in the short run
  - As w varies, the firm will adjust N to maintain equality

Labour Demand Curve in Short Run



- The short run labour demand curve is the MRP curve below ARP
  - Says the amount of labour demanded at every wage
- Slopes down because of diminishing marginal product of labour
  - As more labour is employed, extra amount produced falls
  - Not because of inferior labour, but because more labour is using fixed capital
- As wage falls, firm is willing to use more labour

Demand Curve and Competition in Product Market

- The demand for labour depends on the demand for the product being produced
- It therefore also depends on the structure of the product market
- Firm will always maximize profit by choosing labour where $MRP_N = w$
- But marginal revenue in perfectly competitive market and monopolistic market differ
  - In perfect competition $MR = p$, but in monopoly $MR < p$
  - In perfect competition MR does not fall with more N, but in monopoly MR falls as N increases
- So with less competition in product market
  - Labour demand is lower
  - It falls faster with more N because MR falls as N increases

Demand in the Long Run

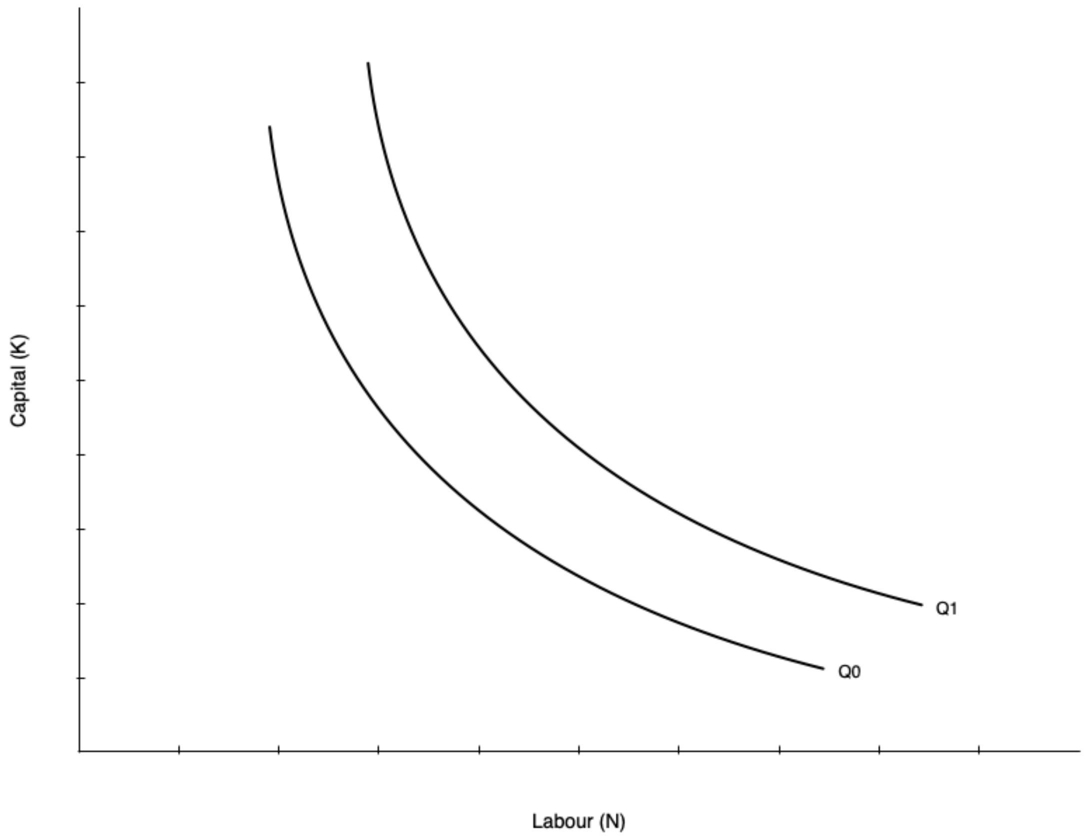
Introduction

- The long run differs from the short run because all inputs are variable
  - Firms can adjust capital as well as labour
- This makes the decision more complicated
  - We now have to consider subsitution between labour and capital
  - We also have to consider economies of scale
- Approach to find optimum occurs in two steps
  - Minimize cost holding output constant
  - Choose output to maximize profit given cost function

Cost Minimization

- First step is to minimize cost of producing given output Q_0
- This occurs when an isoquant is tangent to an isocost line
  - **Isoquant** shows combinations of K and N that produce Q_0
  - **Isocost** line shows combinations of K and N that cost the same amount
  - Analogous to utility maximization with indifference curves and budget constraints
- Assume the cost of capital is r and cost of labour is w
  - Total cost is $C = rK + wN$
- The prouction function is $Q = F(K, N)$
  - We want to minimize C subject to $Q = Q_0$

Cost Minimization



- Graph on left shows two isoquants
  - Combinations of capital and labour that produce the same quantity
- Shape of isoquant is given by the production function
  - Shown as convex, but could be different shapes
  - A linear isoquant would imply perfect substitutes
  - An L-shaped isoquant would imply perfect complements
- Slope of isoquant is the marginal rate of technical substitution (MRTS)
  - MRTS is the rate at which you can

Cost Minimization

- The slope of the isoquant is given by the MRTS

$$MRTS = \frac{MP_N}{MP_K}$$

- Tells you amount of capital you can give up for one more unit of labour holding output constant
- Isoquants are typically display diminishing MRTS
  - When a lot of capital is used, you can give up a lot of capital for one more unit of labour
  - When a lot of labour is used, you can give up only a little capital for one more unit of labour

Cost Minimization

- The other part of cost minimization is the isocost line
  - Gives combinations of K and N that cost the same amount
- Total cost is given by

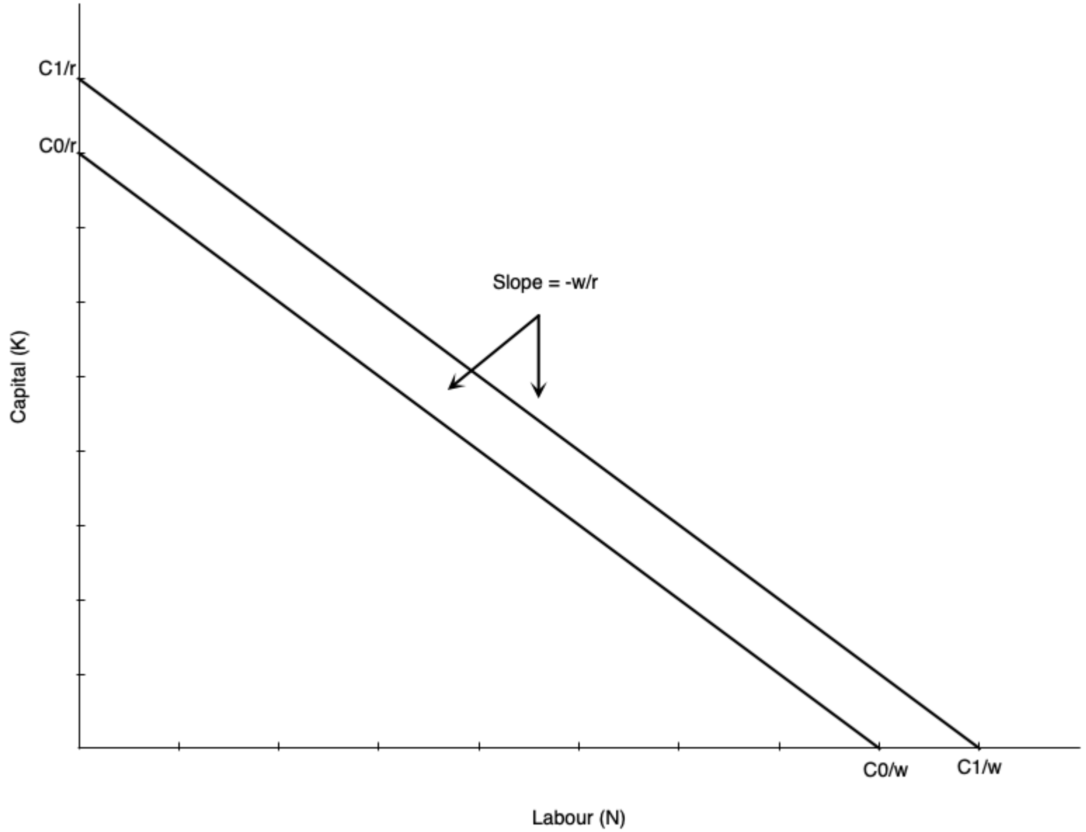
$$C = rK + wN$$

- Rearranging gives the isocost line equation:

$$K = \frac{C}{r} - \frac{w}{r}N$$

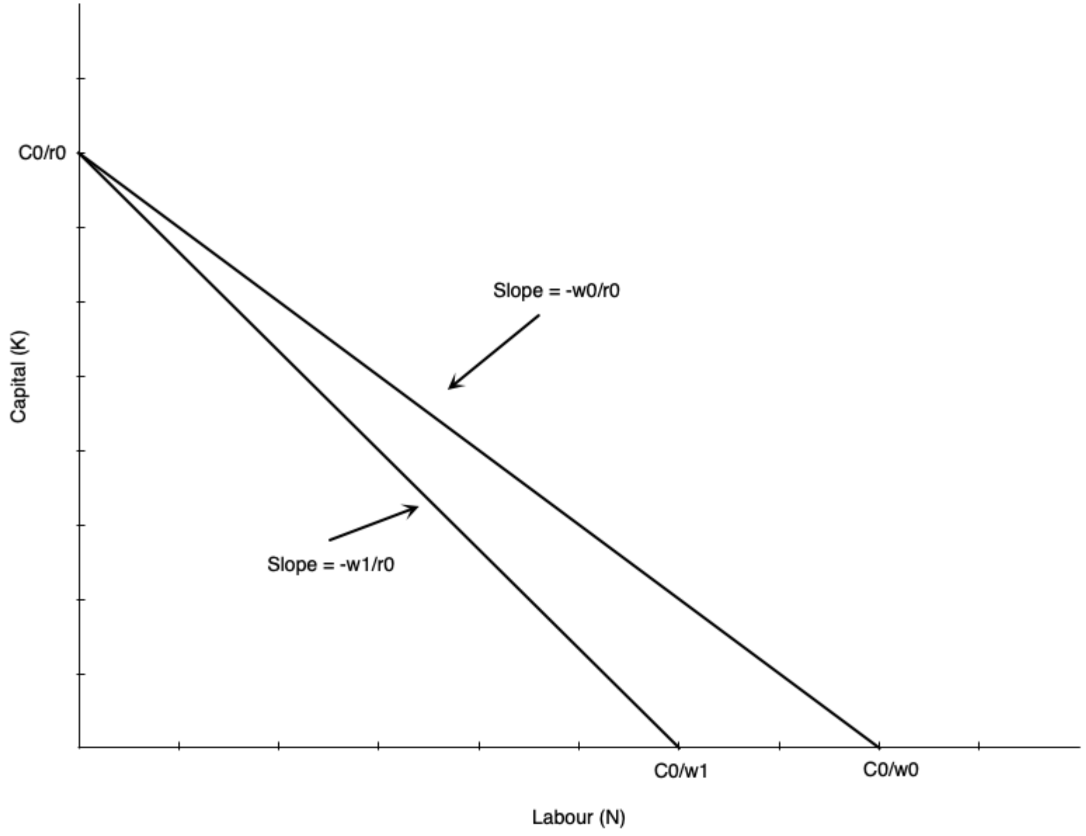
- Slope of isocost line is given by the ratio of input prices
  - As w increases, isocost line becomes steeper
  - As r increases, isocost line becomes flatter

Cost Minimization



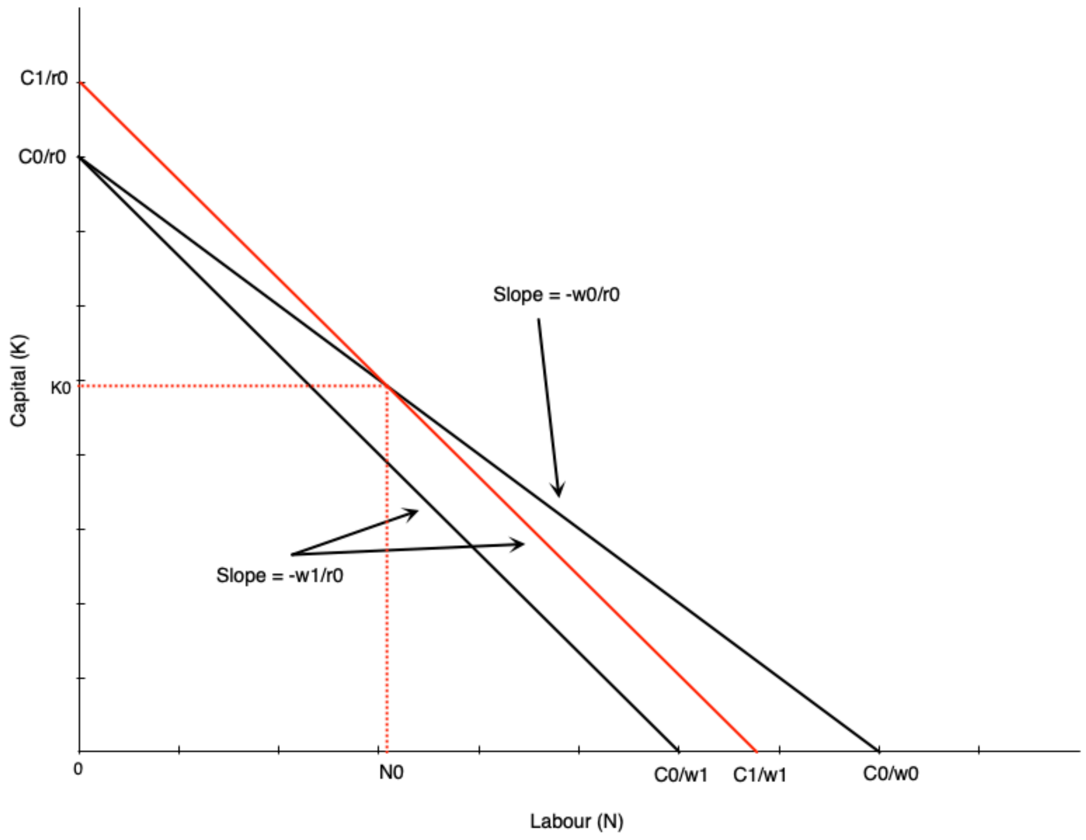
- Graph shows an example isocost
  - Slope is given by ratio of input prices $-w/r$
- If a firm uses only labour, it is equal to C/w
- If a firm uses only capital, it is equal to C/r
- As labour falls by one unit, capital must rise by w/r units to keep cost constant
- Higher cost line shifts isocost line outward (e.g. when C increases to C')

Cost Minimization



- Suppose the wage rises from w_0 to w_1
- Imagine first that we hold cost constant at C_0
- Isocost line would pivot inward
  - Steeper slope as w/r increases
  - If firm uses only capital, vertical intercept remains the same at C_0/r
  - If firm uses only labour, horizontal intercept falls to C_0/w_1
- All combinations of new isocost line lie below old isocost line

Cost Minimization



- To use any of the old combinations of K and N, the firm must increase cost
  - Shifts isocost outward to C_1
  - Magnitude of shift depends on which combination is chosen
- One example depicted to the left
- To use combination (N_0, K_0) at new prices the cost curve shifts out

Cost Minimization

- The cost minimizing bundle is where the isoquant is tangent to the isocost line
- In mathematical terms this means

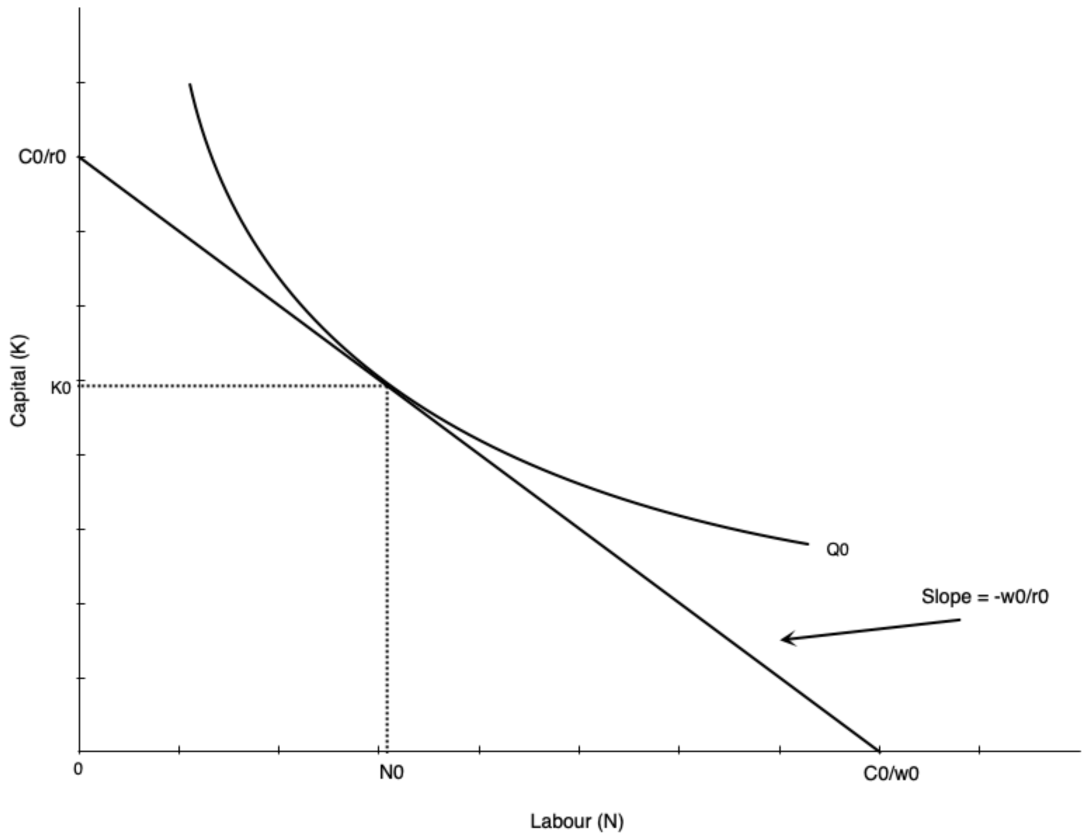
$$\frac{MP_N}{MP_K} = \frac{w}{r}$$

- Says that MRTS is equal to the ratio of input prices
- Alternatively, says that

$$\frac{MP_N}{w} = \frac{MP_K}{r}$$

- The last dollar spent on each input must yield the same marginal product

Cost Minimization



- Cost minimization depicted to the left
- If we hold output constant at Q_0, find the optimal combination of K and N
- Happens at the tangency point between isoquant and isocost line

Cost Minimization

- Note that we have only done cost minimization here
  - Lowest cost given output level Q_0
- Profit maximization involves choosing output level as well
  - Choose Q to maximize profit given cost function
- The second step of choosing output is not depicted here
- If the wage rate goes up, the firm will re-optimize
  - Firm marginal and average cost curves shift up
  - Each firm reduces output
  - Market price rises
- The isoquants at the profit max will shift inward

Cost Minimization – Math

- Suppose production exhibits **decreasing returns to scale**:

$$y = F(K,N) = K^{\alpha}N^{\beta}, \qquad \alpha > 0,\ \beta > 0,\ \alpha + \beta < 1$$

- Input prices: r (capital), w (labour)
- The cost-minimization problem:

$$\min_{K,N}\ C = rK + wN \quad \text{s.t.} \quad K^{\alpha}N^{\beta} \geq y$$

- Lagrangian:

$$\square = rK + wN + \lambda\left(y - K^{\alpha}N^{\beta}\right)$$

Cost Minimization – Math

- First-order conditions:

$$\frac{\partial \square}{\partial K} = r - \lambda \alpha K^{\alpha-1} N^{\beta} = 0$$

$$\frac{\partial \square}{\partial N} = w - \lambda \beta K^{\alpha} N^{\beta-1} = 0$$

$$y = K^{\alpha} N^{\beta}$$

- Divide the first two conditions to eliminate λ:

$$\frac{\beta}{\alpha} \frac{K}{N} = \frac{w}{r}$$

- This is the familiar result:

$$\frac{MP_N}{MP_K} = \frac{w}{r}$$

Cost Minimization – Math

- From production $y = K^{\alpha} N^{\beta}$:

$$N = \left(\frac{y}{K^{\alpha}} \right)^{\frac{1}{\beta}}$$

- Then:

$$\frac{K}{N} = K \left(\frac{K^{\alpha}}{y} \right)^{1/\beta} = K^{\frac{\alpha+\beta}{\beta}} y^{-1/\beta}$$

Cost Minimization – Math

- Plug into $\frac{\beta}{\alpha}\frac{K}{N} = \frac{w}{r}$

$$\frac{\beta}{\alpha} K^{\frac{\alpha+\beta}{\beta}} y^{-1/\beta} = \frac{w}{r}$$

$$K^{\frac{\alpha+\beta}{\beta}} = \frac{\alpha}{\beta}\frac{w}{r} y^{1/\beta}$$

- Solving for the optimal K gives

$$K^*(y; w, r) = y^{\frac{1}{\alpha+\beta}} \left(\frac{\alpha\, w}{\beta\, r}\right)^{\frac{\beta}{\alpha+\beta}}$$

Cost Minimization – Math

- Use $\frac{K}{N} = \frac{\alpha w}{\beta r}$ to get N^*:

$$N^*(y; w, r) = \frac{\beta r}{\alpha w} K^* = y^{\frac{1}{\alpha+\beta}} \left(\frac{\alpha w}{\beta r}\right)^{-\frac{\alpha}{\alpha+\beta}}$$

- The cost function is:

$$C(w, r, y) = rK^*(y; w, r) + wN^*(y; w, r)$$

$$= \phi(w, r)\, y^{\frac{1}{\alpha+\beta}}$$

- Where

$$\phi(w, r) = r\left(\frac{\alpha w}{\beta r}\right)^{\frac{\beta}{\alpha+\beta}} + w\left(\frac{\alpha w}{\beta r}\right)^{-\frac{\alpha}{\alpha+\beta}}$$

Cost Minimization – Math

- Stage 2 is choosing the output level to maximize profit
- Suppose the firm is a price taker in the product market with price p
- Profit:

$$\pi(y) = py - C(y; w, r) = py - \phi(w, r)\, y^{\frac{1}{\alpha+\beta}}$$

- Optimize profit with respect to y

$$\frac{d\pi}{dy} = p - \phi(w, r) \cdot \frac{1}{\alpha+\beta} \cdot y^{\frac{1}{\alpha+\beta} - 1}$$

- The optimal output level is found by setting $\frac{d\pi}{dy} = 0$

$$^{*}(\quad) \quad \left((\alpha+\beta)\, p \right)^{\frac{\alpha+\beta}{1-(\alpha+\beta)}}$$

Cost Minimization – Math

- You can then plug in to get the optimal inputs

$$K^*(w,r,p) = \left(\frac{(\alpha+\beta)\,p}{\phi(w,r)}\right)^{\frac{1}{1-(\alpha+\beta)}} \left(\frac{\alpha\,w}{\beta\,r}\right)^{\frac{\beta}{\alpha+\beta}}$$

$$N^*(w,r,p) = \left(\frac{(\alpha+\beta)\,p}{\phi(w,r)}\right)^{\frac{1}{1-(\alpha+\beta)}} \left(\frac{\alpha\,w}{\beta\,r}\right)^{-\frac{\alpha}{\alpha+\beta}}$$

Cost Minimization – Math

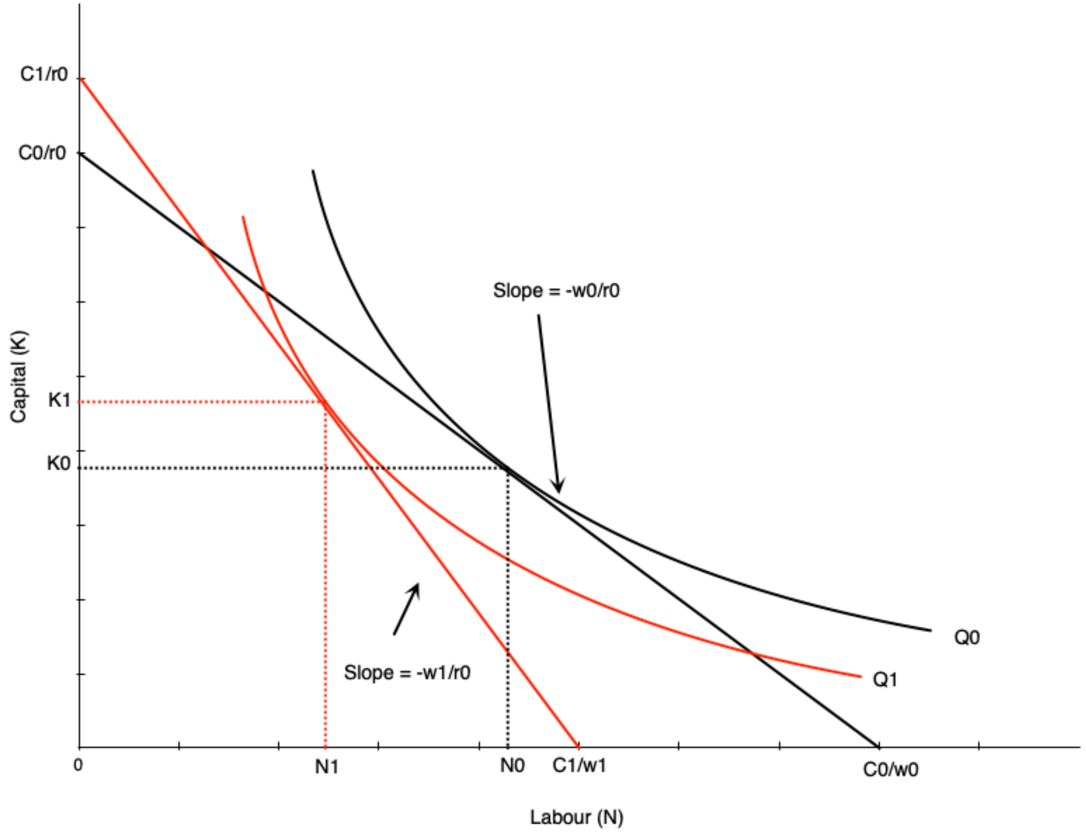
- Do not get too bogged down in the math
  - I will not ask you to recreate this derivation on exams
- The key points are
  - First minimize cost, which gives optimal inputs as function of output and input prices
    - So as you vary any of those three things, the inputs change
  - But given cost function, choose output to maximize profit
    - Output depends on product price and input prices
    - So changes in input prices change both optimal inputs and output level

Deriving Labour Demand in Long Run

Deriving Labour Demand

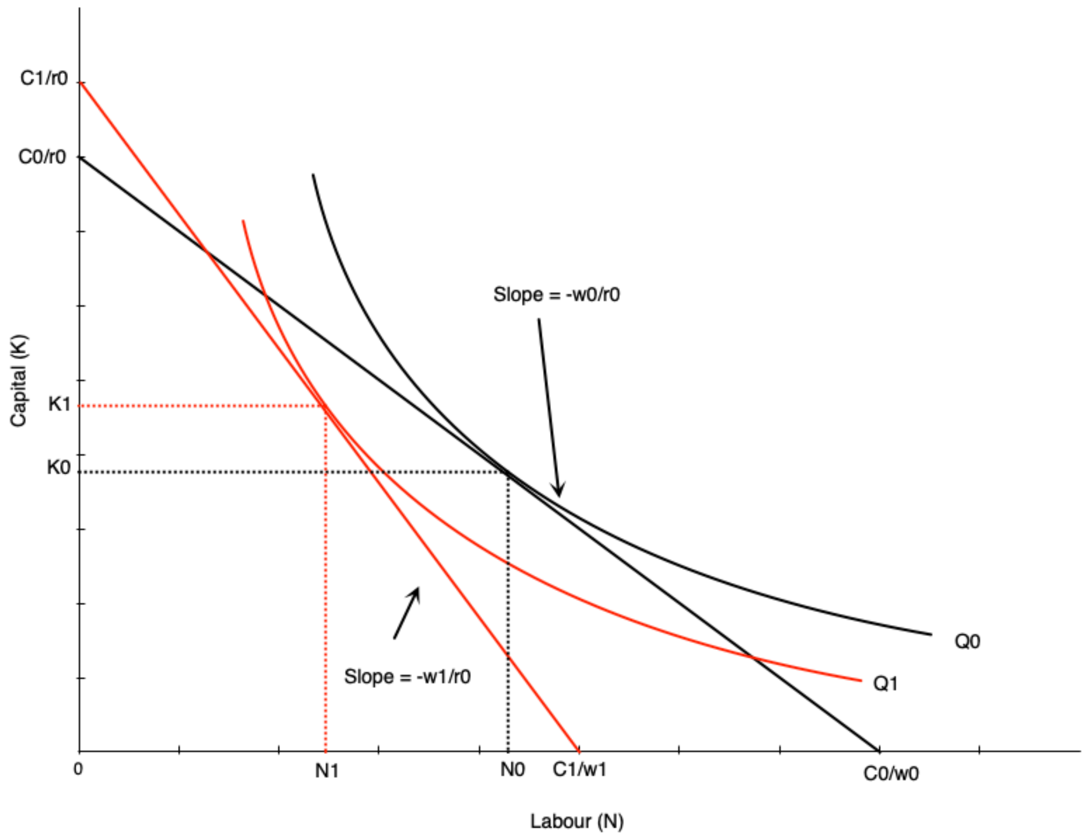
- The labour demand curve plots combinations of w and N such that the firm is maximizing profit
- To derive it, we consider changing wage from w_0 to w_1
  - Isocosts get steeper as w/r increases
  - Firm finds cost minimizing combination of K and N at new wage for each level of output
  - Because cost increases, firm reduces output (optimal isoquant is to the left of original one)
- At new optimum we have
  - Lower quantity of labour demanded
  - Higher wage

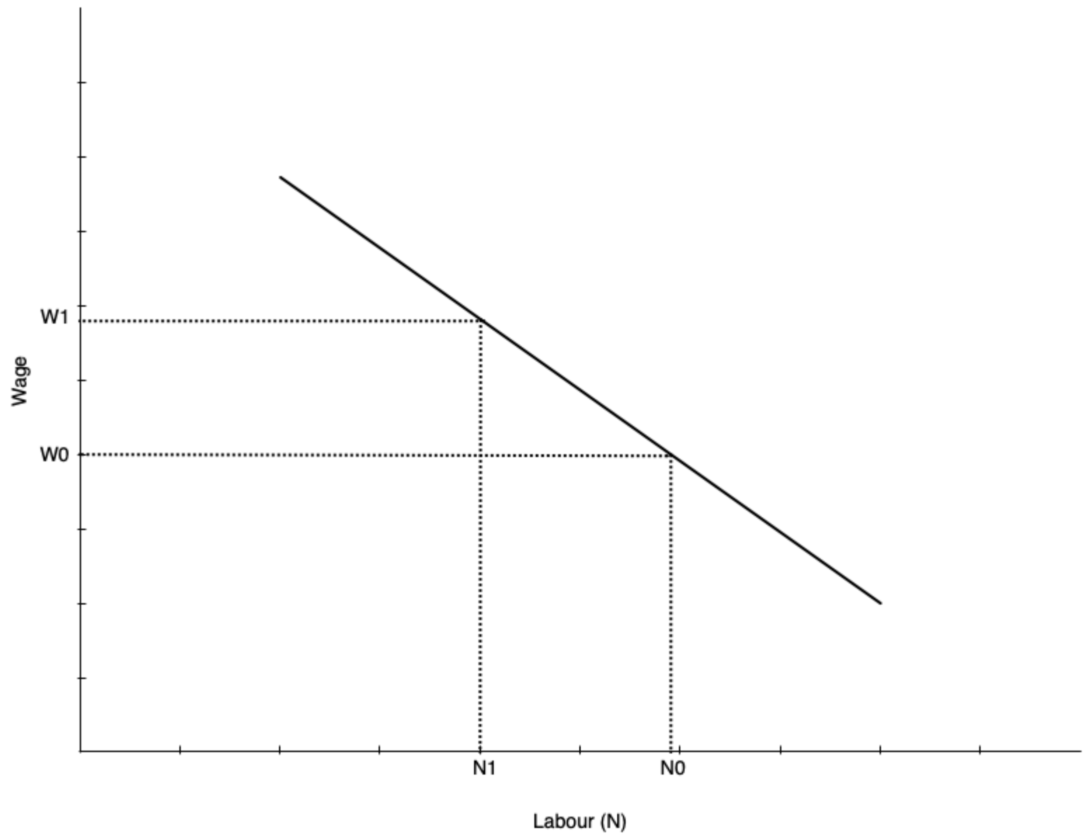
Deriving Labour Demand



- Effect of wage increase on labour depicted to the left
- Wage goes up, which steepens isocost line
- Firm scales down production to lower isoquant
- The tangency between new isoquant and new isocost line shows new optimal labour and capital
- This involves less labour employed
- Effect on capital depends on substitutability between labour and capital
  - The shape of the isoquant

Deriving Labour Demand





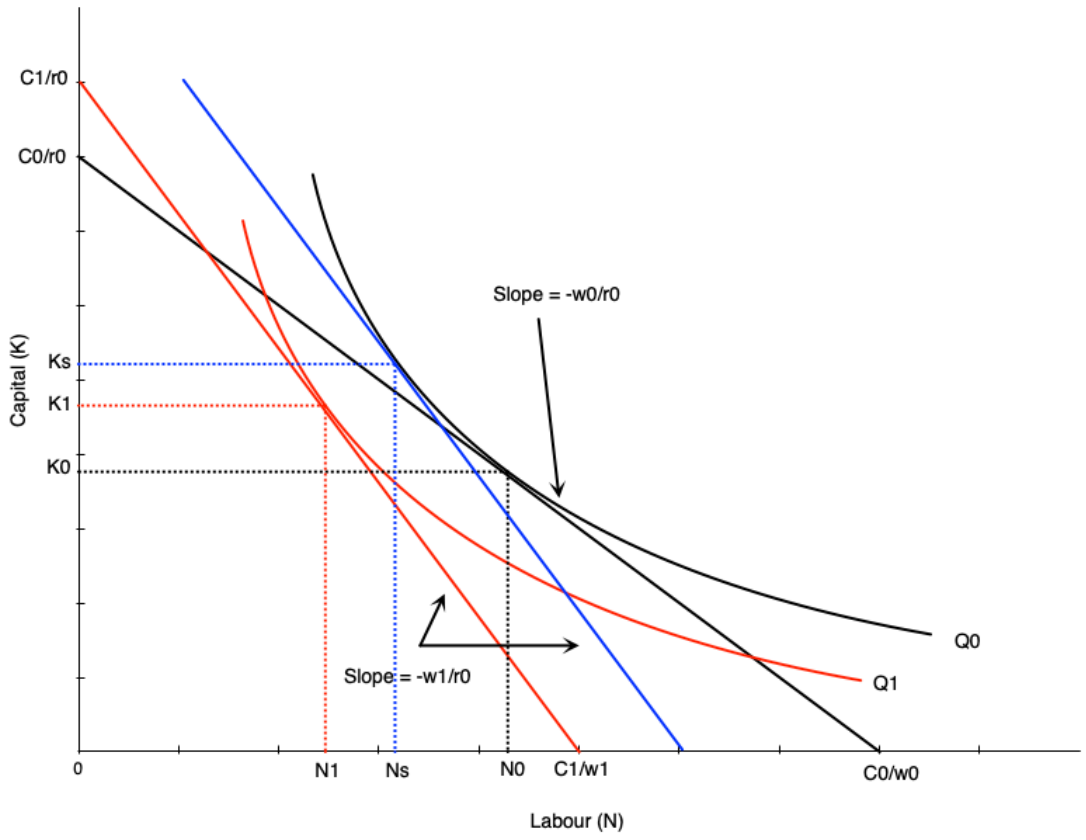
Deriving Labour Demand

- The labour demand curve summarizes the optimal choices of labour at different wage rates
- Graph in previous slide shows to two points on the labour demand curve
  - (w_0, N_0) and (w_1, N_1)
- To derive the full demand curve you can vary the wage rate to other levels
- Recall that the demand curve shows the amount of labour demanded at every wage *all else equal*
  - Other determinants of demand will shift the curve

Scale and Substitution Effects

- When the wage changes, two effects occur that influence labour demand
  - **Substitution effect:** change in input mix due to change in relative input prices
  - **Scale effect:** change in output due to change in cost of production
- The substitution effect makes firms want to substitute capital in place of labour
  - Leads to less labour demanded when wage rises
  - More capital demanded
- The scale effect comes from reduction in output, and leads to less labour demanded when wage rises
  - As production becomes more expensive, firms produce less
  - Use less labour and capital
- For labour demand, scale and substitution effect work in the same direction
  - But opposite directions for capital

Scale and Substitution Effects



- To find substitution effect, find cost minimizing bundle at new wage and old output
  - Blue line in the graph
  - Scale effect is $N_0 \to N_s$
- Scale effect is the change from optimum at new wage and old output to new wage and new output
  - Shift from blue to red line in graph
  - Scale effect is $N_s \to N_1$

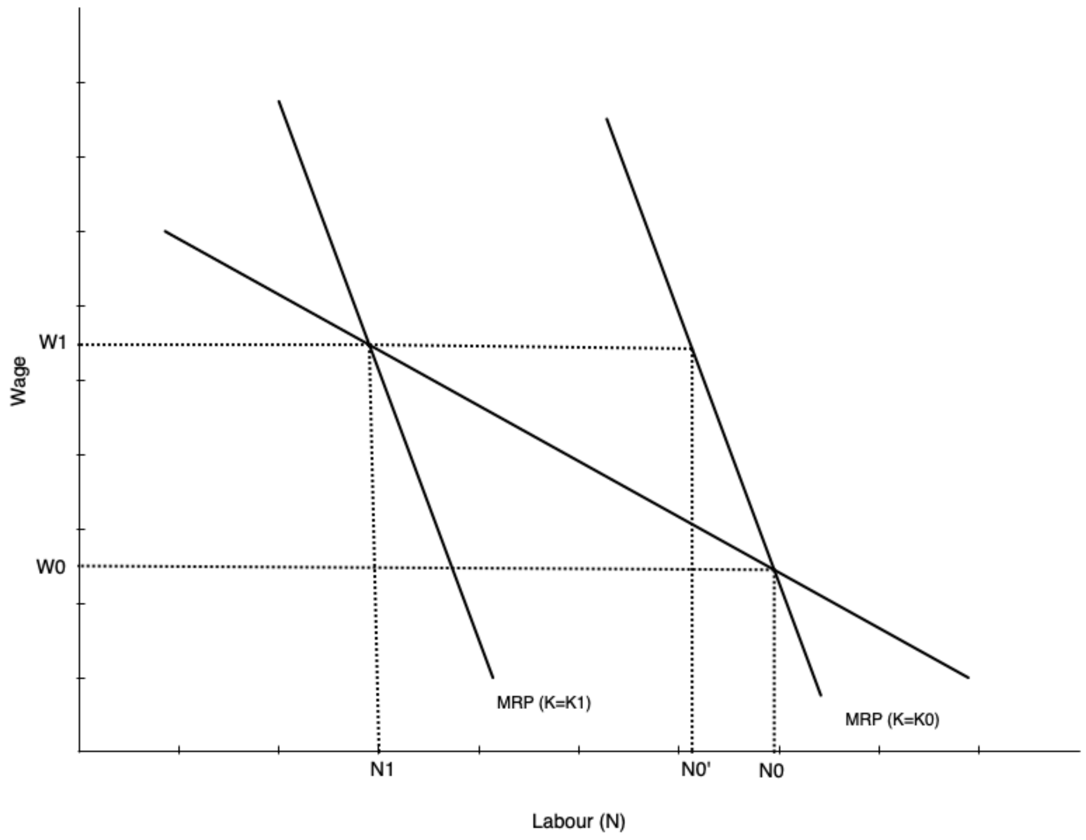
Scale and Substitution Effects

- Notice that the scale and substitution effects of a wage change have opposing effects on capital
- When wage rises, substitution effect leads to more capital being used
  - Firms substitute capital for labour
- But scale effect leads to less capital being used
  - Firms reduce output, so use less of all inputs
- Effect on capital is ambiguous

Relationship to Short-Run Demand

- In the short run, capital is fixed
  - So there is no substitution effect
- In the long run, both substitution and scale effects occur
- Therefore, long-run labour demand is more elastic than short-run labour demand
  - Firms can adjust all inputs in the long run
  - More responsive to wage changes

Relationship to Short-Run Demand



- Graph shows short-run and long-run labour demand curves
- Start with capital fixed at K_0
- Wage goes from w_0 to w_1
- In short run move along SR demand curve from N_0 to $N_{\prime}$
- In long run firm substitutes capital for labour
- Short run curve shifts to the left
- Equilibrium is at intersection of wage and new short run demand

Elasticity of Demand for Labour

Elasticity of Demand

- Recall that elasticity of demand measures responsiveness of quantity demanded to changes in price
- In this case, responsiveness of labour demanded to changes in wage rate
- An elastic labour demand curve means a proportionately larger change in quantity of labour demanded than change in wage
  - Curve is flatter
- An inelastic labour demand curve means a proportionately smaller change in quantity of labour demanded than change in wage
  - Curve is steeper

Factors Affecting Elasticity

- Several factors affect the elasticity of labour demand
  - Ease of substitution between labour and other inputs
    - More or easier substitutes means more elastic demand
  - Elasticity of supply of other inputs
    - Inelastic supply of othe inputs makes it harder to substitute, means less elastic labour demand
    - Happens because other inputs become more expensive when firm tries to substitute
  - Elasticity of demand for the final product
    - Output responding more to price changes means more elastic labour demand
  - Labour share in total costs
    - Higher labour cost share means more elastic labour demand
    - Operates via scale effect

Evidence on Elasticity

- In the USA, evidence suggests elasticity is between -0.15 and -0.75
  - So relatively inelastic
  - Elasticities have increased over time
  - In the long run elasticities are larger
- In Canada, evidence says elasticity is between -0.1 and -0.6
  - Depends on geography and industry
- Key result is that studies confirm that labour demand does slope down

Globalization and Offshoring

Introduction

- Over time Canadian firms have faced increasingly globalized
- Has lead to concerns over what happens to Canadian workers
  - People worry firms will move their operations to other countries with lower wages
  - And that this will destroy Canadian jobs
  - This is a clear concern in the USA right now
- The effect of globalization on jobs is always a concern when trade agreements are reached
  - Manufacturing is especially prominent
  - Currently the automotive industry is a focus of concern

Effect of Trade on Single Market

- Consider first global competition with a specific product
- Increased global competition lowers the price a product can sell for globally
- In a competitive market, this reduces the value of the marginal product
  - Recall that $VMP = p \times MP_N$
- Shifts the demand for Canadian labour inward

Effect of Trade on Single Market

- Another approach is to consider Canadian and foreign labour as two inputs to production
  - Just like we did with capital and labour before
- The optimal choice of Canadian and foreign labour occurs where

$$\frac{MP_{N_C}}{MP_{N_F}} = \frac{w_C}{w_F}$$

- The marginal product per dollar spent on a foreign and Canadian worker are the same

Effect of Trade on Single Market

- What effect does a fall in foreign wages have on Canadian labour demand?
- Two effects occur
  - Substitution effect: firms substitute foreign workers for Canadian workers
    - Reduces demand for Canadian workers
  - Scale effect: lower costs lead to more production
    - Increases demand for Canadian workers
- So overall effect is ambigiuous
  - If they are close subsitutes, substitution effect dominates and less Canadian labour is demanded
  - If scale effect dominates, there could be more Canadian labour demanded

Effect of Trade on Single Market

- Cost is not the only factor
- Productivity is also important
  - If Canadian workers are more productive, will still want to hire them
- How productive are Canadian workers compared to foreign workers?
- Table on next slide shows international comparisons for OECD countries
  - Also changes over time in productivity, compensation, unit labour costs

Effect of Trade on Single Market

TABLE 5.1 Productivity,[1] 2000, and Changes in Productivity, Hourly Compensation,[2] and Unit Labour Costs,[3] 2000–2018 (various OECD Countries)

Country	Productivity in 2000 (as % of Canada)	% Change in Productivity	% Change in Compensation	% Change in Unit Labour Costs
Korea	45	99	32	–23
Mexico	46	2	–44	
New Zealand	79	21	76	42
Japan	87	20	–38	–48
Australia	99	24	45	22
Canada	**100**	18	16	1
Spain	102	17	18	–1
United Kingdom	111	18	–5	–18
Finland	116	20	24	5
Italy	119	1	14	13
Sweden	121	28	17	–7
United States	123	30	2	–19
Germany	126	19	9	–6
France	129	18	19	6
Netherlands	135	15	19	3
Denmark	135	22	33	10
Switzerland	136	19	40	19
Belgium	143	14	21	5
Norway	164	16	46	27
Average	111	22	18	2

NOTES:

1. Output per hour. In descending order, from high to low, in terms of 2000 productivity
2. Defined as the sum of gross wages and salaries and employers' social security contributions. All figures are on an hours-worked basis and are exchange-rate-adjusted and therefore reflect exchange rates and compensation costs expressed in home country currency.
3. Hourly compensation per unit of output (i.e., adjusted for productivity).

SOURCE: Author's calculations based on Organisation for Economic Co-operation and Development, "OECD Productivity Statistics". Available online, accessed August 2020, https://www.oecd.org/sdd/productivity-stats/.

Effect of Trade on Single Market

FIGURE 5.8 Relative Trends in Labour Costs and Productivity: Canada Relative to the United States (base year 1989)

This figure shows relative trends in productivity and labour costs in Canada and the United States. Each point shows the ratio of a cost (or productivity) index in Canada to that in the United States, with 1989 as the base year. For example, in 2012 unit labour costs were 25 percent higher in Canada than in 1989, compared to the United States. Note that these figures do not compare the levels of costs between Canada and the United States, only their trends (relative to 1989).

NOTES: Points represent the relative value of indices of Canadian to U.S. labour costs and productivity (as defined in the notes in Table 5.1). The base year is 1989 (index = 100 for both Canada and the United States).

SOURCE: Author's calculations based on Organisation for Economic Co-operation and Development, "OECD Productivity Statistics". Available online, accessed August 2020, https://www.oecd.org/sdd/productivity-stats/.

Effect of Trade Across Sectors

- Focusing on a single sector misses key points
  - Trade openness affects the entire economy
  - There are adjustments across sectors due to trade
- Recall from EC120 that countries specialize in production of goods where they have a comparative advantage
  - Expands the consumption possibilities of countries
- In a globalized market, countries will allocate more resources towards their comparative advantage sectors
  - Labour will move from non-comparative advantage sectors to comparative advantage sectors

Evidence on Trade and Labour Demand

- There are many trade agreements with other countries
- Most prominent in Canada are NAFTA and more recently CUSMA
- These present opportunities to study the effects of trade on labour demand
- Evidence shows that
  - About 10% of Canadian jobs lost in early 90s were due to free trade
  - Productivity grew faster in manufacturing industries affected by trade

Non-Trade Demand Factors

- At the same time trade is changing, the Canadian labour market is changing
- Over the long term, jobs have shifted away from manufacturing
- Professional, managerial, and administrative service jobs hae been increasing
- Flex or “gig” jobs have been increasing
- All changing for various reasons
  - Technology
  - Consumer preferences
  - Demographics